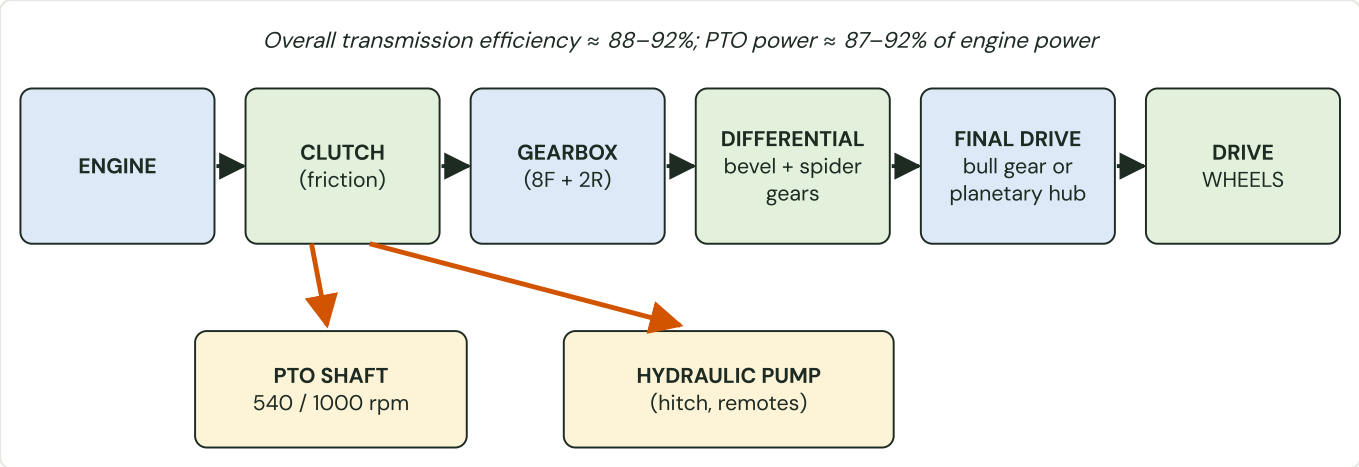
UNIT 4 · TOPIC 5(A)

Transmission Overview and the Clutch

Syllabus topic: Tractor power transmission: clutch, gearbox, differential and final drives



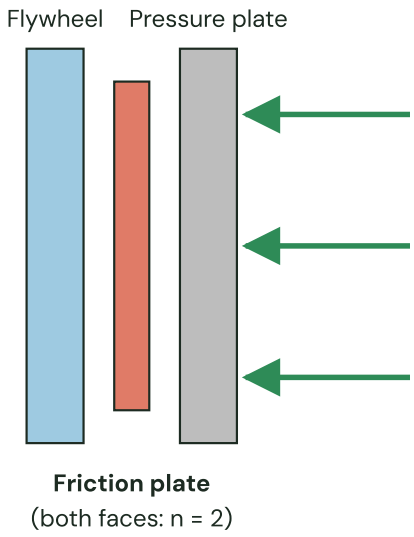
Tractor power transmission and power outlets (redrawn from the source figure)

What each unit does

Unit	Purpose
Clutch	Connect/disconnect the drive smoothly
Gearbox	Torque–speed ratios for 2–30+ km/h from a nearly constant-speed engine
Differential	Turn the drive 90°; allow wheel-speed difference on turns
Final drive	Final torque multiplication near the wheels
Power outlets	Export power via PTO, hydraulics and drawbar (Topic 6)

- Overall transmission efficiency ≈ **88–92%** (each gear pair ~98%).

The clutch



Torque capacity

$T = n \cdot \mu \cdot W \cdot R_m$

$R_m = (Ro + Ri)/2$ (uniform wear)

Ex.: n = 2, μ = 0.3, W = 3000 N,
Ro = 150 mm, Ri = 100 mm → R_m = 125 mm
T = 2 × 0.3 × 3000 × 0.125 = 225 N·m
(> engine torque 195 N·m → no slip)

Dual clutch: two plates — drive + live PTO

Single-plate dry friction clutch (redrawn from the source figure)

Tractor standard: single-plate DRY friction clutch — flywheel + friction plate (both faces work, n = 2) + spring-loaded pressure plate, released by the pedal via the release bearing.

Torque capacity $T = n \cdot \mu \cdot W \cdot R_m$, uniform-wear mean radius $R_m = (Ro + Ri)/2$

- Capacity is designed at **1.25–1.5×** engine torque: it never slips at full load, yet slips FIRST under shock (protecting the gears).

Example. μ = 0.3, W = 3000 N, Ro = 150 mm, Ri = 100 mm, single plate → R_m = (0.15 + 0.10)/2 = 0.125 m; T = 2 × 0.3 × 3000 × 0.125 = **225 N·m**.

Clutch type	Feature
DUAL (two-stage)	Two friction plates on one pedal — first stage stops the ground drive, second stops the PTO → LIVE PTO
Wet multi-plate (oil-immersed)	Heavier / powershift tractors

Adjustment

- Free pedal play: ~**20–25 mm**.
- Too little → slipping, burnt plate. Too much → dragging, hard gear shifts.
- Never ride the clutch.

Practice MCQs

Q106. The correct tractor power flow is:

(a) Engine→gearbox→clutch→diff

(b) Engine→clutch→gearbox→differential→final drive→wheels

(c) Engine→diff→clutch

(d) Engine→final drive→gearbox

Q107. Overall tractor transmission efficiency is about:

(a) 60–70% (b) 75–80%
(c) 88–92% (d) 100%

Q108. Torque capacity of a plate clutch is:

(a) μWR_m (b) n·μ·W·R_m
(c) W/μR (d) nWR/μ

Q109. For a single-plate clutch with both faces active, n =

(a) 1 (b) 2
(c) 4 (d) 0.5

Q110. μ = 0.3, W = 3000 N, Ro = 150 mm, Ri = 100 mm, single plate: T ≈

(a) 112 (b) 168
(c) 225 (d) 450 N·m

Q111. Clutch capacity is designed at about:

(a) 0.8× (b) 1.0×
(c) 1.25–1.5× (d) 3× engine torque

Q112. Free pedal play of a tractor clutch is about:

(a) 2–5 mm (b) 20–25 mm
(c) 60–80 mm (d) 100 mm

Q113. Insufficient free play causes:

(a) Hard steering (b) Clutch slip and burnt friction plate
(c) More power (d) Diff lock

Q114. A DUAL clutch on tractors provides:

(a) Four-wheel drive (b) Live PTO (ground drive and PTO released in two stages)
(c) Better brakes (d) Higher speed

Answers

Q106 — (b) Engine→clutch→gearbox→differential→final drive→wheels
Q107 — (c) 88–92%
Q108 — (b) n·μ·W·R_m
Q109 — (b) 2
Q110 — (c) 225 — *R_m = (0.15 + 0.10)/2 = 0.125 m; T = 2 × 0.3 × 3000 × 0.125 = 225 N·m.*
Q111 — (c) 1.25–1.5×
Q112 — (b) 20–25 mm
Q113 — (b) Clutch slip and burnt friction plate
Q114 — (b) Live PTO (ground drive and PTO released in two stages)

Sources: Liljedahl et al. — Tractors and their Power Units; Jagdishwar Sahay; IS 4931/ISO 500 (PTO), IS 4468/ISO 730 (linkage); NPTEL farm power lectures.